VIVEK PANDHARE

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Education

Michigan State University **Expected May 2028**

Bachelor of Science in Computer Science, GPA: 4.00/4.00 *East Lansing, Michigan*

Relevant Coursework: Data Structures and Algorithms, Systems Programming, Database and Cloud Computing, Discrete Math, Linear Algebra with Computational Applications, Multivariable Calculus, Statistics

Awards: Dean's List (3x)

Experience

Software Engineering Intern **May 2026 – August 2026**

Enginuity Workspace *Troy, MI*

- Implemented **8** REST endpoints in Node.js and Supabase (PostgreSQL) for an AI tutoring platform piloted by **150** students across **5** ECE labs, and rebuilt PDF ingestion for its RAG pipeline with Node pdf-parse.
- Designed a **4**-level hint-escalation policy across **3** subject domains, verified by **44** unit tests, and wrote prompt guardrails against fake citations and off-topic answers that declined **117** of **120** off-topic test questions.
- Fixed **12+** bugs, including an infinite roster-loading loop that blocked every new user from signing up, by tracing failures across the JavaScript frontend and Supabase queries, cutting login time to **under 1 second**.

Artificial Intelligence Engineering Extern **May 2026 – July 2026**

Pfizer Inc. *Remote*

- Developed a Python document-extraction pipeline that uses OpenCV preprocessing and OCR to convert **12,000** pharmaceutical PDFs into structured records validated against JSON schemas for required fields and types.
- Benchmarked PaddleOCR, EasyOCR, and Tesseract, then integrated PaddleOCR into the pipeline after it reached **90%** extraction accuracy, compared with **84%** for EasyOCR and **76%** for Tesseract on the same documents.
- Prototyped a RAG semantic-search service with LlamaIndex, ChromaDB, and open-source LLMs over **160** documents, achieving **87%** Recall@5 with **250 ms** retrieval latency and **3 s** end-to-end responses.

Projects

LLM Gateway | *Python, FastAPI, ChromaDB, sentence-transformers, Docker, Streamlit*

- Built a semantic caching gateway for LLM APIs in Python and FastAPI that answers repeated and reworded questions from ChromaDB vector search instead of calling the model, cutting estimated API cost by **39%**.
- Benchmarked **4** similarity thresholds on **500** paraphrased queries and chose **0.85**, the lowest threshold with **zero** incorrect cache matches, which let the gateway answer **38%** of all benchmark requests directly from the cache.
- Reduced repeat-query latency from ∼**50 s** (local LLM call) to ∼**14 ms**, and added cache expiry, privacy filters for personal and time-sensitive questions, a Streamlit dashboard, and a Docker Compose deployment.

Apex: Low-Latency Trading Engine | *C++20, CMake, GoogleTest, Google Benchmark, Docker*

- Designed a hybrid order book (sorted vector for the top 16 price levels, tree below) with an open-addressing hash map that delivered **1.93x** the throughput of a `std::map` book and cut p99 update latency from **375 to 250 ns** on real data.
- Built a fee-aware smart order router that splits orders across **3** venues simulated from real NASDAQ data and re-routes unfilled shares, lowering execution cost by **42%** versus single-venue routing on AAPL at a **100%** fill rate.
- Rebuilt order books for all **8,892** NASDAQ stocks from **268.7M** ITCH messages with a lock-free multithreaded pipeline, finding the bottleneck by profiling each thread's idle time and sharding it for a **1.73x** end-to-end speedup.

Amazon ReviewGuard | *Python, XGBoost, scikit-learn, spaCy, Pandas, Joblib*

- Trained an XGBoost classifier that detects AI-generated Amazon reviews with **76%** accuracy and **0.85** ROC-AUC on a balanced set of **36,000+** reviews, scored on a held-out test set of **7,278** reviews never seen in training.
- Engineered **18** features with Pandas and spaCy, including length, punctuation, rating, and part-of-speech counts, and tuned the model with a **5-fold** cross-validated grid search over **768** hyperparameter configurations.

Technical Skills

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: FastAPI, Node.js, React, Angular, Streamlit
ML & Data: PyTorch, scikit-learn, XGBoost, Pandas, spaCy, OpenCV, LlamaIndex
Systems & Performance: Multithreading, Lock-Free Data Structures, Profiling, CMake, GoogleTest, Google Benchmark
Databases & Tools: PostgreSQL, Supabase, MongoDB, ChromaDB, Docker, Git, Linux, AWS, REST APIs, OpenAI API